

EMBERGUARD · REQUIREMENTS

EmberGuard requirements

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	EGD-REQ-001	Version	0.3
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/emberguard		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-09-25	Amish Chadha	TRL 3. Record Amish's decisions (EGD-DDR-001); R8 redefined (pump power out of kit scope), R13 at \$425, R4 defaults decided; status of every requirement from EGD-CAL-001
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, reference house and status against the concept
0.1	2026-09-24	Amish Chadha	Initial scaffold

EmberGuard requirements

These requirements are proposals for review, not user-validated needs. Version 0.3 records Amish's decisions of 2026-09-25 (EGD-DDR-001): the budget in R13 is \$425 (D1), R8 now covers the kit's own battery and makes grid-independent water an installation precondition (D6), and the arming defaults in R4 are decided (D7). The status column comes from the TRL 3 calculations in EGD-CAL-001 v0.1 and docs/04-calcs/results.csv . Five requirements are **not met** (R1, R2, R3, R6 and R13), two are **at risk** (R8 and R10), one is not verifiable at TRL 3 (R11) and five are met. The misses in R1 and R3 come from the viewing geometry: the gutter interiors cannot be seen from the ridge-line head.

Table 1. Requirements.

ID	Requirement	Target	Verification	Status at TRL 3 (EGD-CAL-001)
R1	Detect a smouldering ignition on a watched gutter or roof edge	A 100 cm ² (15.5 in ²) hot spot at 300 °C or hotter, anywhere along the watched gutters from 1.5 m to 13 m from the mast, within 10 s	Radiometric calculation from sensor field of view and noise; later a heated-target trial	Not met: gutter interiors are hidden behind the eave; a flat spot on the roof edge gives 2.8 K at 8.2 m, below the 5 K trigger; a hot debris face of 100 x 50 mm is seen to 26 m (EGD-CAL-001, A4, B9 to B11)
R2	Detect single landed embers	A 10 mm glowing ember at 600 °C or hotter within 8 m of the mast, within 10 s	Same as R1	Not met: 2.4 K at 8 m centred in a pixel; reaches the 5 K trigger within 5.5 m (B4, B6)
R3	Watch both roof planes	Both eave gutters of the reference house, each from 1.5 m from the gable end to the far end	Line-of-sight check on the parametric model	Not met: roof edges in view from 1.25 m to 13.15 m; gutter interiors not visible (A3, A4)
R4	Arm only in fire weather	Automatic arming when sustained wind is 30 km/h (19 mph) or more, or gusts 50 km/h (31 mph) or more, with relative humidity 20 % or less for 10 min; manual and remote arming; defaults adjustable and reviewed with a local fire agency (decided, D7)	Design review of the arming logic	Met by design
R5	Start water quickly	Water at the farthest head within 60 s of a confirmed detection	Line fill calculation	Met: 55 s for the longer zone, filling the detection-side zone first (C3)
R6	Wet the gutter and roof edge	Net water on a 1.2 m wide strip along each eave (gutter, fascia and first metre of roof) of 5 mm/h or more, averaged over each spray cycle, at the design wind of 30 km/h	Droplet drift screening model; later a spray trial in wind	Not met: leeward eave 0.8 mm/h at 8.3 m/s and 2.6 mm/h at 4.2 m/s; windward eave 7.4 mm/h (C8)
R7	Use little water	Steady draw 4 L/min (1.1 US gal/min) or less; 1,000 L (264 US gal) or less for a 4 h ember event	Flow calculation from head ratings and duty cycle	Met: 4 L/min and 965 L, 3.5 % margin (C5)
R8	Work through a grid outage (redefined, D6)	The kit battery alone runs sensing, logic and valves for 72 h armed plus 4 h of spraying, with no solar input. Water supply that does not depend on grid power (gravity feed, generator or battery pump) is an installation precondition provided by the homeowner and is outside the kit	Energy budget; installation check of the water supply	At risk: 61.0 Wh needed; 69.1 Wh usable at 25 °C (met), but 62.2 Wh at 0 °C and 55.3 Wh at end of life leave little or no margin (D4)
R9	Fail safely	Loss of sensor data while armed starts spraying (decided, D8); low battery, no water pressure or a faulty valve raises an alarm; power loss closes the normally closed valves (decided, D9)	Failure modes review	Met by design

ID	Requirement	Target	Verification	Status at TRL 3 (EGD-CAL-001)
R10	Survive fire weather until the front arrives	Mast and head survive gusts of 120 km/h (75 mph); electronics operate from -10 to 60 °C ambient; head shaded from sun and radiant heat	Wind load and thermal calculations	At risk: mast 27 MPa (factor 8.9) and standoffs factor 3.5 met; the enclosure reaches about 71 °C in sun at 60 °C ambient, above the battery's limit; radiant heat not estimated (E3, E4, F1)
R11	Install without roof work or mains wiring	Mast, cable and spray lines fixed to walls, fascia and gutter lips with clamps; no roof penetrations; 12 V DC only; installable by two people with hand tools in 6 h or less	Design review; later a timed trial	Not verifiable at TRL 3: met by design except the install time
R12	Tell people what it is doing	Local siren and status light; phone alert when armed, spraying, faulted or low on battery when a network is available; event log of detections, wind and valve actions	Design review	Met by design; phone alerts need a working network
R13	Stay within the concept budget	Kit parts \$425 or less (budget raised from \$300, D1), excluding the water source and pump, and excluding priced options	Priced BOM	Not met: \$571 (G1)

■ Reference house and assumptions

- **House.** Single storey, 12 x 8 m (39 x 26 ft) footprint, 2.7 m walls, gable roof at 22 degrees with the ridge along the long side, 500 mm overhangs, gutters along both long eaves (13 m each, 26 m total). The sensor mast stands off one gable end at the ridge line.
- **Water.** A 1,000 L tank with a pump on its own supply (decided, D5). The supply must hold about 2.3 bar at the valve manifold at 4 L/min (EGD-CAL-001, C4). EmberGuard does not include the tank or pump.
- **Ember event.** 4 h of active spraying within a 72 h armed period. This is a design assumption, not a measured duration; real ember exposure varies from minutes to many hours.
- **Heads.** Six micro-sprinklers per eave at 2.2 m spacing, about 40 L/h each at 2 bar (typical catalogue figure for small irrigation heads, to be confirmed with a named part).
- **Wind drift.** Estimated at TRL 3 with a droplet screening model (EGD-CAL-001, section C) in place of the TRL 2 guess of 50 %. The leeward eave is the worst case. The model is uncertain and needs a spray trial.
- **Ember targets.** Emissivity 0.9, sensor band 8 to 14 µm, background 300 K. See EGD-CAL-001, section B.

Safety: EmberGuard does not replace evacuation. Requirements R4 to R9 describe behaviour while the house is empty. Nothing in these requirements permits anyone to stay behind to operate or watch the system.